The Berry ratio in the transfer pricing: concept, economic logic and Jurisprudential development

1. Introduction and Historical foundations

1.1. Origin of berry ratio:

In transfer pricing, transactions between **associated enterprises** (for example, between a parent company and its subsidiary) are compared with transactions between **independent enterprises** in order to determine whether the pricing reflects the **Arm's Length Principle (ALP)**. For this purpose, five principal transfer pricing methods are recognised. Three of these—**Comparable Uncontrolled Price Method (CUP), Resale Price Method (RPM), and Cost Plus Method (CPM)**—are considered traditional transaction methods. The remaining two—**Transactional Net Margin Method (TNMM)** and the **Profit Split Method (PSM)**—are profit-based methods.

Within the framework of profit-based analysis, various **Profit Level Indicators (PLIs)** are used to evaluate whether a transaction yields an arm's length outcome. The **Berry Ratio** is one such PLI, particularly relevant in cases involving distribution or intermediary functions.

The Berry Ratio was first introduced by **Charles Berry** in the landmark **1979 case of *E.I. DuPont de Nemours & Co. v. United States***. The case concerned the multinational chemical company DuPont, whose parent company was located in the United States and which operated a subsidiary in Switzerland known as **DISA**. DISA purchased chemical products from the U.S. parent and resold them to other DuPont subsidiaries located across Europe and other regions, as well as to third-party customers.

In this arrangement, DISA functioned essentially as a **low-risk distributor**. It purchased products from the parent company and resold them onward, retaining approximately **20% of the profits** generated from these transactions. Due to the unusually high profitability reported by DISA, the **Internal Revenue Service (IRS)** issued a deficiency notice and appointed **Charles Berry as an independent expert** to analyse the pricing structure.

Because DuPont operated in a **highly specialised chemical industry**, it was difficult to identify reliable comparable companies with similar product margins. As a result, Berry adopted an alternative analytical approach. Instead of focusing on product margins, he decomposed the **gross profit** of the distributor into functional components such as **marketing, advertising, and distribution activities**. He then identified independent European businesses engaged in comparable functional activities and analysed their financial performance. Berry calculated the relationship between **gross profit and operating expenses**, thereby developing what later came to be known as the **Berry Ratio**.

For the years **1959 and 1960**, the Berry Ratio for DISA was calculated at **281.5 and 397.1**, respectively, whereas comparable independent companies showed ratios of only **108.3 and 129.3**. Based on this substantial disparity, the court observed that **no independent enterprise had achieved such extraordinarily high ratios within the**

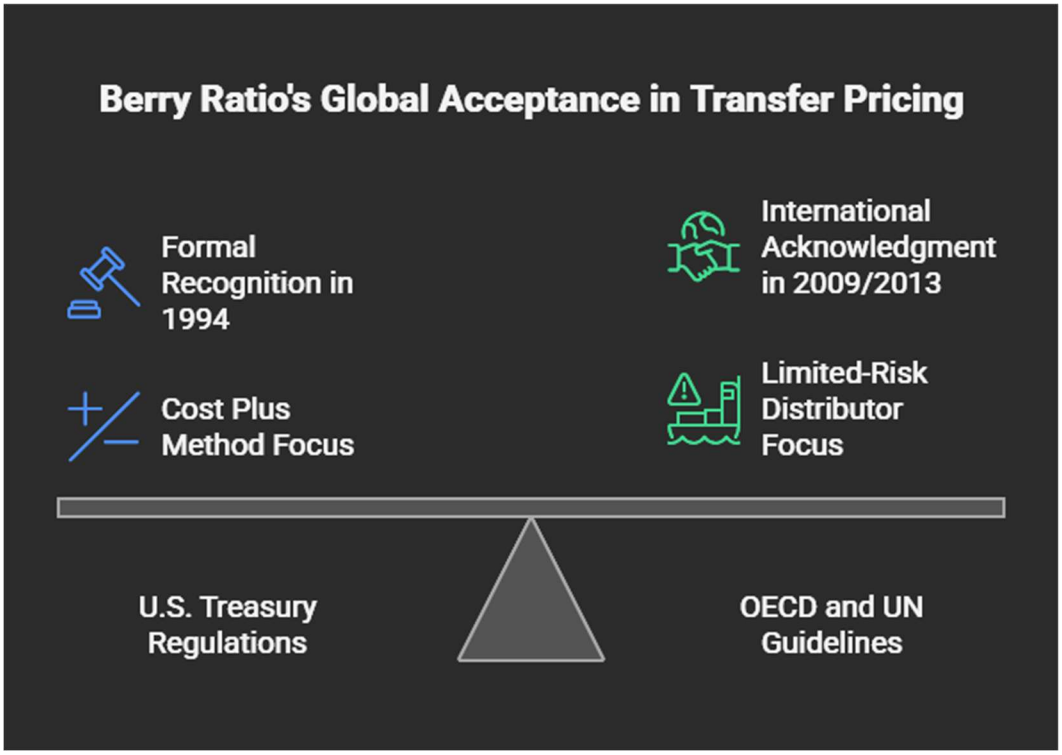
preceding hundred years, indicating that the pricing arrangement did not reflect arm's length conditions.

1.2. Regulatory evaluation

After the **DuPont case**, the Berry Ratio gradually gained recognition in transfer pricing regulations. It was first discussed in the **1988 U.S. Treasury White Paper**, which analysed the DuPont decision and considered the use of the Berry Ratio as a financial indicator for determining arm's length outcomes.

Later, the **1994 U.S. Treasury Regulations under Section 482** formally recognised the Berry Ratio as a financial ratio that may be used as a **Profit Level Indicator (PLI)**, particularly while applying the **Cost Plus Method (CPM)**.

At the international level, the **OECD Transfer Pricing Guidelines** first referred to the Berry Ratio in **2009**, and the **UN Practical Manual on Transfer Pricing** also acknowledged its use in **2013**, mainly in cases involving limited-risk distributors or intermediary functions where operating expenses are the primary value driver.



2. Mechanics and Economic Logic

2.1. The Core Formula:

Berry ratio defined as:

$$Berry\ Ratio = \frac{Gross\ Profit}{Operating\ Expenses}$$

Where:

Gross Profit (GP) = Sales – Cost of Goods Sold

Operating Expenses (OE) = Selling, general, and administrative expenses related to distribution activities

Because:

Gross Profit = Operating Profit + Operating Expenses Gross

the ratio can be rewritten as:

$$BR = 1 + \frac{Operating\ Profit}{Operating\ Expenses}$$

Thus, the Berry Ratio effectively measures how much profit is generated for each unit of operating expense incurred.

2.2. Practical example of Berry ratio:

Assume a controlled entity has Net Sales ("NS") of $100 and Cost of Goods Sold ("COGS") of $30. Its Gross Profit ("GP"), which is NS minus COGS, is therefore $70. If its Operating Expenses ("OE") are $50, then its Operating Profit ("OP"), which is GP minus OE, is $20. The Berry Ratio ("BR") is defined as GP/OE, which in this example is $70/$50 = 1.4. See the calculations in Table 1 below.

Table 1: Calculation of the Berry Ratio

Net Sales	$100
Cost of Goods Sold	$30
Gross Profit	$70
Operating Expenses	$50
Operating Profit	$20
Berry Ratio	1.4

Since Gross Profit can be decomposed into Operating Profit plus Operating Expenses, the Berry Ratio can also be expressed as:

BR = GP / OE = (OP + OE) / OE = OP / OE + 1

Applying the same formula to the example:

BR = OP / OE + 1 = 20 / 50 + 1 = 1.4

Both calculations therefore produce the same result. The first expression (GP/OE) resembles a gross markup on operating costs, while the second **(OP/OE + 1) reflects a net markup on operating costs.**

This structure explains why the Berry Ratio can function as a Profit Level Indicator under CPM or TNMM. Conceptually, it resembles the Cost Plus Method, because it evaluates the remuneration of a distributor by reference to the cost of the services performed.

For a distributor that merely purchases finished goods and resells them without modification, **gross profit functions similarly** to revenue, while **operating expenses represent the cost of performing distribution services**. Operating expenses therefore serve as a proxy for the extent of the services performed, such as marketing, sales, and distribution.

Under the arm's length principle, the analytical question becomes: **what markup would an independent enterprise earn for performing the same distribution services?** The Berry Ratio answers this by measuring the **appropriate markup** on operating expenses, which represent the value-added services provided by the distributor.

2.3. Linking to Capital Intensity:

The Berry Ratio can also be analysed in relation to the capital intensity of a firm. Starting from the earlier expression:

BR = 1 + OP / OE

This can be rearranged by incorporating Operating Assets (OA):

BR = 1 + [OP / OE × OA / OA] = 1 + [OP / OA × OA / OE]

In this equation:

OP / OA represents the **Return on Capital Employed (ROCE)**, measuring profitability relative to operating assets.

OA / OE represents the **asset intensity** of the firm.

Accordingly, the Berry Ratio can be interpreted as:

Berry Ratio = 1 + (ROCE × Asset Intensity)

Thus, the Berry Ratio reflects not only profitability but also the capital intensity of the enterprise.

Illustration

Assume two firms earn the same ROCE of 10%:

- Firm 1: OA = $200 million, OE = $20 million
- Firm 2: OA = $100 million, OE = $20 million

Berry Ratios:

- Firm 1: BR = 1 + (0.10 × 200/20) = 2.0

- Firm 2: BR = 1 + (0.10 × 100/20) = 1.5

Although both firms earn the same return on capital, Firm 1 reports a higher Berry Ratio because it is more asset-intensive.

Why the Berry Ratio Becomes Unreliable

The Berry Ratio assumes that operating expenses are the primary driver of value creation. This assumption holds for limited-risk distributors or service providers where functions such as marketing and distribution generate the value.

However, where firms differ significantly in asset intensity, the ratio becomes unreliable because:

- Returns generated by assets are not captured, as the ratio focuses only on operating expenses.

- Comparability is distorted, since firms with higher operating assets may show higher Berry Ratios despite having the same profitability.

- Profit drivers are misidentified, as profits attributable to capital or intangibles may be incorrectly attributed to operating expenses.

Therefore, when OA/OE ratios vary significantly across firms, the Berry Ratio may not provide a reliable Profit Level Indicator, unless adjustments are made for differences in asset intensity

3. Situations Appropriate for Application

3.1. General consideration for application of Berry ratio:

First, the **Section 482 Regulations, OECD Guidelines, and the UN Transfer Pricing Manual** recognise the Berry Ratio as a **Profit Level Indicator for one-sided transfer pricing methods**, particularly the **Cost Plus Method (CPM)** and the **Transactional Net Margin Method (TNMM)**. One-sided methods are suitable where the **tested party performs routine functions, does not own valuable intangibles, and does not assume significant entrepreneurial risks**. If the tested party performs entrepreneurial functions or owns valuable intangibles—such as brands, patents, or trade secrets—the Berry Ratio becomes inappropriate because the value created cannot be captured merely through operating expenses.

Second, the **denominator of the Berry Ratio (Operating Expenses)** must represent the **extent of services performed by the tested party**. Since the ratio measures returns relative to operating costs, OE must reflect only the **value-adding activities undertaken by the entity**.

Third, certain items normally included in operating expenses should be **excluded** if they do not reflect services performed by the entity. In particular:

- **Interest expense** should be excluded because it reflects the firm's financing structure rather than operational activity.
- **Depreciation** should generally be excluded because it relates to the timing of asset acquisition rather than current service functions.

However, Charles Berry noted a qualification: if **controlled and uncontrolled entities have similar relationships between operating expenses and capital expenditure**, depreciation may already be implicitly reflected in the margins earned on operating expenses.

Fourth, the appropriateness of the Berry Ratio depends on the **nature of activities performed**. If a distributor performs **multiple functions**, the Berry Ratio may still be used provided the functions can be **segmented and separately priced**. This approach was applied in the **DuPont case**, where Berry separately analysed marketing, advertising, and distribution functions.

However, the Berry Ratio is **generally unsuitable for manufacturing activities**, as manufacturing businesses tend to be **capital intensive**. Where an entity performs both manufacturing and distribution, the ratio should only be applied if the activities can be **clearly segmented**.

Fifth, **comparability across industries and geographic markets** presents challenges. The level and structure of operating expenses vary significantly across industries. For example, the operating expense structure of a **pharmaceutical distributor** may differ substantially from that of a **commodity or automobile distributor**. Differences may also arise due to **economies of scale, market share, intensity of functions, and strategic positioning**, as well as regional cost variations.

Another important issue concerns **pass-through costs**. Operating expenses used in the Berry Ratio should include only costs related to **services performed by the tested party**. Costs relating to **materials purchased from third parties or pass-through expenses** should be excluded. For example, in the case of an **advertising agency**, purchases of media time from television or newspapers should not form part of operating expenses, since those services are provided by the media companies rather than the agency itself.

A related requirement is that **benchmark companies must be treated consistently**. Pass-through costs must also be removed from the operating expenses of **uncontrolled comparables**, and the classification of **COGS and operating expenses must be consistent** between the tested party and comparables. Otherwise, the Berry Ratio becomes unreliable.

These considerations led Charles Berry to observe that although **the concept of the Berry Ratio is simple, its practical application can become complex**. When applied under appropriate circumstances and with careful adjustments, it can provide a useful method for determining an **arm's length outcome for certain distribution or intermediary transactions**. Outside those specific conditions, however, the Berry Ratio may not provide a reliable measure of arm's length pricing.

3.2. Comparison between the Berry ratio and ROS (Return on sales)

The Berry Ratio is often used as a secondary Profit Level Indicator (PLI) for controlled distributors, particularly where the traditional indicator—Return on Sales (ROS)—is not reliable or where additional confirmation is required. Both the OECD Transfer Pricing Guidelines and the UN Transfer Pricing Manual provide guidance on the circumstances in which the Berry Ratio may be preferred over ROS.

Conditions for Using the Berry Ratio

According to the OECD Guidelines, the Berry Ratio is appropriate only when the following conditions are satisfied:

- The value of the functions performed (taking into account assets used and risks assumed) is proportional to operating expenses.
- The value of the functions performed is not proportional to sales.
- The tested party does not perform other significant functions, such as manufacturing, that should be remunerated using another method or indicator.

The UN Transfer Pricing Manual reiterates these conditions and explains that the Berry Ratio measures the return to the value-adding functions of the entity, assuming that the costs of these functions are reflected in operating expenses. Therefore, the ratio is most suitable for selling and marketing entities or service providers, provided that the functions and expenses are properly classified and the uncontrolled comparables are functionally similar and do not own significant

intangible assets. Correct classification between Cost of Goods Sold (COGS) and Operating Expenses (OE) is essential for comparability.

Key Limitations Identified by the UN Manual

The UN Manual highlights several limitations in the application of the Berry Ratio:

- Sensitivity to cost classification, particularly the distinction between COGS and operating expenses.
- Exclusion of costs related to maintaining intangible property, which may distort the measurement of value creation.
- Reduced reliability when asset intensity increases, since the ratio focuses primarily on operating expenses.

Relationship Between the Berry Ratio and Return on Sales

Both the OECD Guidelines and the UN Manual emphasize that the Berry Ratio should not be used when the value of functions performed is proportional to sales. In such situations, the Berry Ratio provides no additional information beyond Return on Sales.

Assume operating expenses are proportional to net sales:

$$OE = \alpha \times NS, \text{ where } \alpha \text{ is a constant.}$$

Since the Berry Ratio is defined as:

$$BR = OP / OE + 1$$

Substituting OE:

$$BR = OP / (\alpha NS) + 1$$

Given that Return on Sales (ROS) = OP / NS, the relationship becomes:

$$BR = ROS / \alpha + 1$$

or equivalently:

$$ROS = \alpha (BR - 1)$$

This relationship shows that when operating expenses are a fixed proportion of sales, the Berry Ratio and Return on Sales move in perfect correlation.

Illustration

Assume:

$\alpha = 0.3$

$ROS = 0.20$

Then:

$$BR = 0.20 / 0.30 + 1 = 1.667$$

If ROS doubles to 0.40:

BR = 0.40 / 0.30 + 1 = 2.333

Thus, changes in ROS are exactly mirrored by changes in the Berry Ratio, demonstrating that when operating expenses are proportional to sales, the Berry Ratio does not provide additional analytical insight beyond ROS.

3.3. Effect of Firm Size: Berry Ratio vs. Return on Sales

One advantage of the **Berry Ratio**, particularly for **limited-risk distributors (LRDs)**, is that it is less sensitive to differences in the **size of comparable companies**. As a distributor's **net sales increase**, its **operating expenses (especially SG&A)** generally do not increase proportionally because of **economies of scale**. Consequently, changes in sales volume may significantly affect **Return on Sales (ROS)** but have a much smaller effect on the **Berry Ratio**.

Illustration

Consider three distributors with similar **functions, assets, and risk profiles**, but different scales of sales.

Table 2: How Changing Net Sales Affects the Berry Ratio and ROS

	Firm A	Firm B	Firm C
Net Sales (NS)	$100	$1,000	$10,000
Cost of Goods Sold (COGS)	$90	$900	$9,000
Gross Profit (GP)	$10	$100	$1,000
Operating Expenses (OE)	$6	$30	$150
Operating Profit (OP)	$4	$70	$850
Berry Ratio (BR)	1.667	8.333	41.667
Return on Sales (ROS)	0.04	0.07	0.085

In this example:

Firm B's sales are 10 times Firm A's, and **Firm C's sales are 10 times Firm B's**.

COGS and gross profit scale proportionally with sales, but **operating expenses increase only fivefold**, reflecting **economies of scale in SG&A costs**.

The **ROS ratios remain relatively close**, ranging from **4% to 8.5%**, while the **Berry Ratios differ substantially** across the firms. The example illustrates that when ROS is used as the PLI, **larger distributors with higher sales may appear to earn significantly higher profits**, even though the **functional and risk profiles are similar**.

Because the Berry Ratio focuses on **returns relative to operating expenses rather than sales**, it can provide a **more appropriate benchmark when comparing firms with substantially different sales volumes**.

Relevance for High-Volume, Low-Margin Businesses

The Berry Ratio is therefore often considered more suitable for **trading houses and distributors engaged in high-volume, low-margin activities**. In such businesses, even a **small ROS percentage applied to very large sales volumes can generate large operating profits**, which may not reflect the **actual value-adding activities performed by the distributor**.

In these situations, the Berry Ratio better captures the relationship between **operating expenses and the services performed**, particularly where **pricing or sales volumes are determined elsewhere within the multinational group**. Many transfer pricing cases applying the Berry Ratio have involved **Japanese *sōgō shōsha* trading companies**, which typically operate as **limited-risk distributors dealing in high-volume, low-margin products**.

4. Accounting Classification Issues and Reliability of the Berry Ratio

Under **U.S. Section 482 Regulations** and the **IRS APA Study Guide**, the reliability of the **Berry Ratio** depends on the **comparability of operating expenses between the tested party and uncontrolled comparables**. Operating expenses are generally defined as **SG&A expenses**, which may include **depreciation**, but exclude **interest expense, income taxes, and other non-operational costs**. These expenses typically cover **advertising, marketing, sales, warehousing, distribution, and administrative activities**.

However, the Berry Ratio is **highly sensitive to cost classification**. Under **U.S. GAAP**, companies have discretion in allocating costs between **COGS and SG&A**, and service providers often classify certain service costs as **COGS rather than operating expenses**. Financial databases such as **Compustat** may also automatically allocate costs like **transportation and warehousing to COGS**, which can distort the calculation.

This classification issue can significantly affect the Berry Ratio. For example, if **Gross Profit is EUR 12 million and Operating Expenses are EUR 8 million**, the Berry Ratio equals **1.5**. If **EUR 4 million of operating expenses are reclassified as COGS**, operating expenses fall to **EUR 4 million**, increasing the Berry Ratio to **3.0**, even though the underlying activity remains unchanged.

The **OECD Guidelines** acknowledge that the Berry Ratio is **sensitive to cost classification** and may create comparability issues, particularly where **pass-through costs** exist. The problem is more pronounced in **Europe**, where accounting and database practices differ across countries. For example, some databases calculate gross profit using **COGS**, while others rely on **Material Costs**, which include only raw material purchases and not the full range of direct production costs.

The **UN Transfer Pricing Manual** also highlights the risk that **expense classification between operating expenses and COGS may be manipulated**, affecting comparability. It recommends excluding **interest expense and extraneous income**, while the treatment of **depreciation and amortization** depends on the specific facts and circumstances.

Because the Berry Ratio is highly dependent on **consistent classification of operating expenses and COGS**, its proper application requires **careful review of financial statements and detailed benchmarking data** for comparable companies. The UN Manual also lists the Berry Ratio among the financial indicators that tax authorities may review during **preliminary desk audits** in transfer pricing examinations.